

Intro GGT

Homework 3

Question 1. Let (X, d) be a geodesic metric space. We will show the equivalence between the Gromov product hyperbolicity and having slim triangles.

Part I: From hyperbolicity to slim triangles

Assume that (X, d) is δ -hyperbolic. Let $[x, y, z]$ be a geodesic triangle. Let $p \in [x, y]$ and $q \in [x, z]$.

- Using twice the definition of the Gromov product δ -hyperbolicity, prove that

$$(p, q)_x \geq \min\{(p, y)_x, (y, z)_x, (z, q)_x\} - 2\delta.$$

- Show that if the minimum is attained at $(p, y)_x$, then

$$d(p, q) \leq d(x, q) - d(x, p) + 4\delta.$$

Do the analogous computation when the minimum is attained at $(z, q)_x$.

- Deduce that

$$d(p, q) \leq \max\{|d(x, p) - d(x, q)|, d(x, p) + d(x, q) - 2(y, z)_x\} + 4\delta.$$

- Conclude that all geodesic triangles in X are 4δ -thin, hence 4δ -slim.

(Hint: Consider 6 cases: first let $p \in [x, y]$ such that $d(x, p) \leq (y, z)_x$ and choose $q \in [x, z]$ such that $d(x, p) = d(x, q)$.)

Part II: From slim triangles to distance estimates

Assume now that all geodesic triangles in X are δ -slim. Let $x, y, z \in X$, and fix a geodesic $[y, z]$.

- Using the triangle inequality, show that for any $p \in [y, z]$,

$$(y, z)_x \leq \frac{1}{2} \left((d(x, p) + d(p, y)) + (d(x, p) + d(p, z)) - (d(y, p) + d(p, z)) \right) \leq d(x, p).$$

Deduce that

$$(y, z)_x \leq d(x, [y, z]).$$

- Fix geodesics $[x, y]$ and $[x, z]$. Use δ -slimness to show that there exists a point $p \in [y, z]$ such that

$$d(x, y) \geq d(x, p) + d(p, y) - 2\delta, \quad d(x, z) \geq d(x, p) + d(p, z) - 2\delta.$$

- Deduce that

$$(y, z)_x \geq d(x, p) - 2\delta,$$

and conclude that

$$d(x, [y, z]) \leq (y, z)_x + 2\delta.$$

Part III: From slim triangles to hyperbolicity

- Let $x, y, z, t \in X$, and fix geodesics $[x, y]$, $[y, z]$, and $[z, x]$. Use δ -slimness to show that

$$d(t, [x, z]) \geq \min\{d(t, [x, y]), d(t, [y, z])\} - \delta.$$

- Using the estimate from Part II relating distance to a side and the Gromov product, deduce that

$$(x, z)_t \geq \min\{(x, y)_t, (y, z)_t\} - 3\delta.$$

- Conclude that (X, d) is 3δ -hyperbolic.